Topological Analysis of Temporal Trust Structure

A proposed study of the Bitcoin Alpha network

Project idea

Compare temporal windows through Jaccard pseudometrics, persistent homology, and distances between Betti curves.

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SF2956 Topological Data Analysis

Project summary, 2026

Motivation

- Bitcoin Alpha users rated other users from -10 to $+10$.
- Each rating has a direction, a sign, a weight, and a timestamp.
- The timestamps allow us to study how trust behaviour changes over time.

Why a topological approach?

Counts describe network size. TDA describes how groups and higher-order structures persist across distance scales.

ratings $\longrightarrow$ behavioural similarity $\longrightarrow$ multi-scale structure

Bitcoin Alpha dataset

Property	Value
Users	3,783
Ratings	24,186
Rating range	-10 to $+10$
Positive ratings	about 93%

One row of data

(source, target, rating, time)

- directed
- weighted and signed
- temporal

Source: Stanford Network Analysis Project, Bitcoin Alpha trust weighted signed network.

Research question

Main question

How does the multi-scale topological structure of user trust behaviour change over time?

- Do users form groups with similar positive-trust behaviour?
- Do these groups merge differently in early and late windows?
- Do H_0 and H_1 signatures vary across time?
- Can temporal windows be grouped into distinct regimes?

Exploratory claim: clear groups or temporal regimes may not exist.

Temporal windows and preprocessing

1. Convert Unix timestamps to dates.
2. Divide the data into six-month windows.
3. Keep the latest rating for repeated source–target pairs.
4. Define ratings above zero as positive trust.
5. Retain users with at least three positive ratings.
6. Keep at most 200 highly active users per window.

The same rules will be applied to every window.

Course notation

$$I = \{X_d \mid d \in D\}$$

D: temporal windows

X_d: active users in window d

Jaccard pseudometric

For user u in window d, define the positive-trust neighbourhood

$$T_d(u) = \{v \mid u \text{ positively rates } v\}.$$

$$d_d(u, v) = 1 - \frac{|T_d(u) \cap T_d(v)|}{|T_d(u) \cup T_d(v)|}.$$

A small distance means that two users positively rate similar people.

Small example

$$T(A) = \{D, E, F\}$$

$$T(B) = \{D, E, G\}$$

$$d(A, B) = 1 - \frac{2}{4} = 0.5$$

Important: a small distance does not mean that A directly trusts B.

Connection to the course pipeline

Project step	Course concept
Temporal windows	$I = \{X_d \mid d \in D\}$
Jaccard distance	Metric restriction and pseudometric
Groups at scale t	Partitions $X_d/(d_d)_t$
Multi-scale clustering	Single linkage and dendrogram
Geometric construction	Vietoris–Rips filtration
Topological summary	H_0, H_1, barcodes, and Betti curves
Window comparison	l_1 distance and hierarchical clustering

Multi-scale groups and H_0

Single-linkage dendrogram

- At small t, only very similar users merge.
- Increasing t produces larger behavioural groups.
- Merge heights record dissimilarity between groups.

Zero-dimensional homology

- H_0 records connected components.
- A bar dies when one component merges into another.
- A slow decrease of $\beta_0(t)$ indicates dispersed behaviour.

Primary analysis

H_0 will be the main result because it directly describes how behavioural groups merge across scales.

Vietoris–Rips filtration and H_1

At scale t, connect users when $d_d(u, v) \leq t$ and fill every clique with a simplex.

$$\mathrm{VR}_{t_1}(X_d) \subseteq \mathrm{VR}_{t_2}(X_d) \subseteq \cdots, \qquad t_1 \leq t_2.$$

Barcode interpretation

- birth: a feature appears
- death: the feature disappears
- long bar: persistence over a wider scale range

Role of H_1

- detects loops in the similarity space
- serves as a secondary analysis
- weak or short-lived loops remain reportable

H_1 loops are not direct trust cycles in the original network.

Comparing temporal windows

Each window produces Betti curves

$$\beta_k^d(t) = \dim H_k(\mathrm{VR}_t(X_d)), \qquad k \in \{0, 1\}.$$

Compare windows with an l_1 distance

$$D_k(d_i, d_j) = \int_0^1 \left| \widetilde{\beta}_k^{d_i}(t) - \widetilde{\beta}_k^{d_j}(t) \right| dt.$$

Betti curves $\longrightarrow$ distance matrix $\longrightarrow$ window dendrogram

Two different dendrograms

User dendrograms describe structure within a window. The final dendrogram compares the windows themselves.

Planned outputs and interpretation

Outputs

- activity over time
- representative user dendrograms
- H_0 and H_1 barcodes
- Betti curves across windows
- window-distance heatmap
- temporal-window dendrogram

Possible interpretations

- Faster β_0 decrease: users connect at smaller scales.
- Persistent H_1: more complex similarity structure.
- Separate window branches: possible temporal regimes.
- Similar curves: relative stability under this representation.

These are hypotheses to examine, not current findings.

Scope, feasibility, and limitations

Deliberate scope

- one main distance: Jaccard
- one main linkage: single linkage
- homology in dimensions 0 and 1
- at most 200 users per window
- no higher-dimensional computation

Main limitations

- rating strength is reduced to positive or non-positive
- negative ratings do not enter the main distance
- window width and activity threshold affect the output
- Jaccard similarity does not imply direct mutual trust
- the analysis does not establish causality

Proposed project and requested feedback

Summary

The project represents each temporal window as a pseudometric space, computes persistent topological summaries, and clusters windows through distances between their Betti curves.

Questions for discussion

- Is the Jaccard representation appropriate for this project?
- Are six-month windows a reasonable initial choice?
- Is a primary focus on H_0, with H_1 as secondary, sufficient?
- Should we compare raw or normalised Betti curves?

Thank you

Backup: likely questions

Why TDA instead of ordinary community detection?
TDA follows connected components and higher-order structure across all distance thresholds rather than returning only one partition.

Why Jaccard distance?
Each user is represented by a finite set of positively rated users. Jaccard distance directly compares the overlap of these sets.

Why is the user space a pseudometric space?
Two different users may have identical positive-trust neighbourhoods and therefore distance zero.

What if no clear clusters appear?
We will report that the selected representation does not reveal strongly separated behavioural groups or temporal regimes.

Backup: parameter checks

Before the main analysis, we will examine

- ratings and active users per six-month window
- the effect of minimum activity thresholds 2 and 3
- whether the 200-user cap removes a substantial share of activity
- sensitivity to nearby window widths if the default is too sparse
- raw versus normalised β_0 curves

Parameter changes will be documented and applied consistently across windows.

References


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